Gaps and doublets, rational learning, and a knowledge norm on forms*

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Abstract

Inflectional gaps (**forgoed*/**forwent*) and doublets (✓*dived*/✓*dove*) can seem surprising in light of common assumptions about morphology and learning. Perhaps understandably, morphologists have troubled themselves with such cases (especially with gaps) and have offered different ways in which they can be accommodated within the morphological component of the grammar, often by making major departures from common assumptions about morphology and learning. I will suggest that these worries and the proposed remedies are premature. The worries arise from the unmotivated assumption that all gaps and doublets are necessarily derived within individual grammars. Once this assumption is abandoned, the observed properties of gaps and doublets become much less puzzling. The only new assumption that is needed is that speakers only use forms that they know (and not just find it probable) to be correct.

1 Introduction

In morphological inflection it is common to find exactly one possibility for a given combination of features: *You* ✓*rang*/**rought* but *You* **brang*/✓*brought*. Cases in which more than one possibility is available (doublets; ✓*dived*/✓*dove*, ✓*proved*/✓*proven*, ✓*leaped*/✓*leapt*, ✓*dreamed*/✓*dreamt*) or, perhaps worse, none (gaps; **forgoed*/**forwent*, *have* **dived*/**diven*, *have* **strided*/**stridden*) are somewhat unusual. Such cases (for reasons briefly touched upon below, with more emphasis on gaps) have received attention in the generative morphological literature as early as Halle (1973) and Hetzron (1975) and remain an active topic of research today (see Sims 2023 and Pertsova 2024 and references therein).

The interest in such cases is understandable: while gaps and doublets can be represented in principle on common assumptions about morphology, and while such representations can be acquired in principle on common assumptions about learning, the actual properties of

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gaps and doublets raise several worries. I describe two key assumptions about representations and learning in section 2 and review three worries in section 3, followed in section 4 by some remedies that have been proposed in the literature. In section 5 I explain why I think that the worries are premature and why the remedies should wait until there are good reasons to reject what I take to be further aspects of the null hypothesis, namely that: (a) the hypotheses that learners entertain need not be monolingual and can in principle include multiple grammars that might not have external differentiating factors such as register (cf. Kroch 1994 and Roeper 1999); and (b) learners sometimes do not encounter enough evidence to choose a clear winning hypothesis. With multiple grammars, as in (a), otherwise puzzling doublets are predicted. And with incomplete learning, as in (b), gaps follow as long as speakers are required to actually know (rather than just find it probable) that the forms they use are derived within the correct hypothesis.

2 Background assumptions on representations and learning

This section reviews common assumptions about morphology and learning. Section 2.1 reviews the idea that morphology involves competition for specificity. Section 2.2 reviews the idea that learning involves balancing the simplicity of hypotheses with their fit to the data. These assumptions make it possible in principle to state and learn gaps and doublets in certain cases.

2.1 Representations: competition for specificity

A common assumption within morphology is the Pāṇinian principle of competition for specificity: if two forms match a given context and one is more specific than the other, the more specific one wins. Within Distributed Morphology (DM; Halle and Marantz 1993, 1994, and see Bobaljik 2017 for further discussion and references), which I assume here for concreteness, this competition is referred to as the Subset Principle, which says that of the forms that match a given feature specification, the one that matches the most features wins (see Halle 1997, and see Embick and Marantz 2008 and Embick et al. 2023 for further discussion of competition in DM and in other approaches to morphology). For the present tense in English, for example, *-s* (specified for [present, 3sg]) wins over $\emptyset$ (specified just for [present]). Specificity is generally taken to apply not just to features but also to the context of insertion. For the past tense in English, $\emptyset$ (specified for contexts such as $\sqrt{\text{PUT}}$) and $i \rightarrow$ ablaut (specified for contexts such as $\sqrt{\text{SING}}$) win over the default *-ed*.

As noted in the literature, competition for specificity is not inherently incompatible with the representation of gaps and doublets. Gaps can arise from the absence of a default. Consider, for example, the genitive singular form of masculine nouns in Polish, for which Dąbrowska (2001) argues that there is no default. Some forms take a *-a* ending, some other forms take a *-u* ending, but neither ending extends automatically to new forms. Kottum

(1981) and Gorman and Yang (2019) report that Polish speakers are sometimes uncomfortable with either *-a* or *-u*.[1] For example, Kottum (1981) reports that the speakers he consulted with were unsure about the genitive singular of the city name *Tarnobrzeg*, and Gorman and Yang 2019 identify *balon* 'balloon', among several other nouns, as gapped. These observations can be accounted for if for the relevant speakers, neither *-a* nor *-u* is specified for insertion in the context under consideration (e.g., in the context of $\sqrt{\text{BALON}}$ for speakers for whom *balon* is gapped).

In English, the past tense form *-ed* is clearly a default, as seen by *wug*-testing and other considerations, so the account of the *forgo* gap cannot be based on *-ed* not being applicable in the context of $\sqrt{\text{FORGO}}$. But following Arregi and Nevins (2014)'s account of gaps (building, in turn, on Harley 2014), one can say that the expression of the root $\sqrt{\text{FORGO}}$ itself is specified to appear in certain contexts and that these contexts do not include the past tense.

Doublets, meanwhile, can arise from a tie between winners. Returning to genitive singular masculine nouns in Polish, Kottum (1981) reported that *portfel* 'wallet' can take either *-a* or *-u*, which can be accounted for if both options are specified for insertion in the context of $\sqrt{\text{PORTFEL}}$.

In the English case of *dived/dove*, the fact that *-ed* is a default again complicates things. If both the default *-ed* and the more specific ablaut form are compatible with $\sqrt{\text{DIVE}}$, the latter should win and rule out the former. But again this problem can be overcome. As a first step, one could consider a hyper-specific form of *-ed* that realizes the past tense in the context of $\sqrt{\text{DIVE}}$. This first step is not enough, since this form of *-ed* will be more specific than the ablaut realization (which applies also in the context of roots such as $\sqrt{\text{DRIVE}}$ and $\sqrt{\text{STRIVE}}$). But if we now state a hyper-specific ablaut process that applies only in the context of $\sqrt{\text{DIVE}}$, neither of the two hyper-specific forms will be more specific than the other, and the result will be free variation.

2.2 Learning: balanced simplicity

So morphological theory predicts that certain configurations will result in gaps and that certain configurations result in doublets. Can these configurations be learned?

To answer this question we should say more about how learning occurs. A very general perspective sees the child as a scientist, trying to find the best hypothesis within a space of possible hypotheses in light of the observed data. This view was central to early work in Generative Grammar (Chomsky 1957, 1965) and will be assumed here (noting that there are other perspectives on learning as well). Within this view, we should say more about: (a) the hypothesis space, (b) the data, and (c) what makes one hypothesis better than another.[2]

[1] It is not clear to me to what extent the gaps and doublets reported in the earlier literature hold for current speakers of Polish. Several speakers with whom I consulted did not share the reported judgments. I will not attempt to characterize this variation here and will use the description in the literature for illustration only.

[2] This is not an exhaustive list; e.g., the details of how the child traverses the hypothesis space could also affect what is learned from a given input.

Regarding (a), morphological theory amounts to a commitment as to a part of hypotheses that is central here. The precise commitment changes between theories; in DM, it involves vocabulary items (VIs), written in a specific way, that are inserted after syntax, subject to competition for specificity.

For (b), a common assumption is that the child relies primarily on distributional evidence: children observe the forms that are used around them, but cannot rely on systematic corrections, paradigmatic organization, morphological decomposition, or other kinds of help. See Marcus (1993) and Calamaro and Jarosz (2015), among others, for discussion.

As to (c), we need to specify the evaluation metric that tells the child how two hypotheses compare in light of the data. The literature highlights the roles of two factors in comparing hypotheses: the simplicity of the hypotheses, sometimes referred to as *economy*, and the tightness of fit of the hypotheses to the data, sometimes referred to as *restrictiveness*. I will discuss each in turn before concluding (with much previous literature) that both factors are crucial and that they should be balanced against each other.

Economy was the evaluation metric of early generative grammar, e.g., as in Chomsky and Halle (1968). For current purposes it can be stated as follows: if two hypotheses licensed by UG are compatible with the data and one is more complex than the other, reject the more complex one. The appeal of economy is that it leads to generalization: for a simple grammar to be compatible with the data, it cannot simply memorize the data (since memorizing requires a very complex grammar) and needs to generalize.

The problem with economy when used on its own is that it will lead to gaps nowhere and to doublets everywhere. Consider, e.g., how economy would deal with learning the past tense in English. The past-tense VI $\emptyset$ in actual English is specific: it is listed for a handful of roots and is impossible elsewhere. But the correct VI for $\emptyset$, with a list of applicable roots, is more complex to write than a default such as *-ed*. A default $\emptyset$ is compatible with the data: the observed forms can all be generated. Economy will therefore prefer the wrong, overgenerating grammar where both *-ed* and $\emptyset$ are defaults. And when all VIs of the past tense are defaults, speakers will be free to choose for each instance of any root in the context of the past tense which VI to use. In other words, all forms are doublets, an obviously wrong prediction.[3] Economy alone, then, is inadequate: the existence of non-defaults illustrates the known fact that humans generalize in a much more cautious way than what economy alone supports.

Caution is the intuition behind restrictiveness, which tells the learner to prefer hypotheses that fit the data as closely as possible. We can state it as follows: if two hypotheses licensed by UG are compatible with the input data, but one fits the input data more tightly than the other, reject the one that provides the looser fit. Restrictiveness will prevent the learner from acquiring a default $\emptyset$ VI for the past tense in English: a hypothesis with a VI that allows $\emptyset$ just in the context of $\sqrt{\text{PUT}}$ and several other roots provides a tighter fit, so the overly general hypothesis loses. In this way, restrictiveness makes crucial use of indirect negative evidence: it treats absence of evidence as evidence of absence. See Dell (1981)

[3] Similar considerations will ensure that the VIs of roots will never be learned as tense-specific, which rules out the second route mentioned above for deriving gaps.

and Manzini and Wexler (1987) for further discussion of restrictiveness.

The problem with restrictiveness when used on its own is that it is overly conservative: absence of evidence is not always evidence of absence. Since restrictiveness lacks nuance in its inference from indirect negative evidence, it will lead to a surfeit of gaps. For example, if the child has heard *talk* in the present but not yet in the past tense, they will immediately treat the past tense as a gap. This is of course empirically incorrect: humans generalize beyond the data, and *-ed* is readily applied by speakers to nonce forms. So restrictiveness alone is also inadequate.

But having observed the problems of economy and restrictiveness when either is used in isolation, we can now see how each might be able to restrain the other. Economy alone leads to overgeneralization, and restrictiveness alone is overly conservative, but a learner that balances the two will generalize (for purposes of economy) while still trying to maintain a good fit to the data (restrictiveness). Learning a specific $\emptyset$ VI for the past tense in English, with a list of allowable roots ($\sqrt{\text{PUT}}, \dots$) is more complex than a default $\emptyset$, but the costs in economy are more than justified by the gains in restrictiveness. Similarly, if a given form is only ever seen in the present tense, a balanced learner might conclude that the past tense for that root is a gap: stating this by making the root tense-specific will be costly in terms of economy, but at some point the gains in restrictiveness might justify this. Fleshing out this kind of balance requires more formal detail than I will provide here. One common way of implementing it is known as Minimum Description Length (MDL; Rissanen 1978), with mathematical foundations by Solomonoff (1964); it is also very close to Bayesian inference. For discussion and examples of MDL in the context of grammar learning see de Marcken (1996), Goldsmith (2001), Goldwater and Johnson (2004), Katzir (2014), Rasin and Katzir (2016), and Rasin et al. (2021), among others.

3 Worries

I just reviewed why both gaps and doublets are in principle compatible with common assumptions about representations and learning. Still, there are good reasons not to leave things at that.

3.1 Worry 1: Representing and learning actual gaps

Consider what it would take for a balanced learner to conclude that a given form is gapped due to a lack of default. For example, that *forgo* has no past tense because of a VI for $\sqrt{\text{FORGO}}$ that is specific to non-past environments. Such a VI is more complex than a default VI for $\sqrt{\text{FORGO}}$ that is not specified for tense. For a balanced learner, this means that the tense-specific VI would need to have a sufficient advantage in restrictiveness over the default VI if it is to be acquired. The tense-specific VI is indeed more restrictive than the default one, and over a sufficiently large corpus the gains in restrictiveness will outweigh the extra costs of storing the tense-specific entry. But given that *forgo* is a relatively rare verb to begin with, it will take a very large amount of data for the learner to reach this point. This makes an account of the *forgo* gap in terms of a tense-specific VI for $\sqrt{\text{FORGO}}$ unlikely.

The learnability concern does not arise in high-frequency cases. Moreover, motivation for storing a specific VI rather than a simpler default version can come from the ability of the specific VI to account for the absence of a class of forms. In the case of genitive masculine singulars in Polish, for example, the collective absence of several gapped forms might justify the additional storage costs of specific, non-default *-a* and *-u*. Still, cases such as the *forgo* gap need to be accounted for.

As to doublets, the hyper-specific entries discussed earlier for doublets such as *dived/dove* seem unnatural. For example, they would require saying that the *-ed* in *dived* is inherently different from the (shared) *-ed* in *walked* and *talked*. The learning challenge, on the other hand, is potentially easier for doublets than it was for gaps, since for doublets the learner can rely entirely on positive evidence (e.g., hearing both *dived* and *dove*). However, a relatively high number of instances might be needed to justify storing complex, hyper-specific entries rather than treating one element of the doublet as noise.

3.2 Worry 2: The distribution of gaps and doublets

The outline of representations and learning above says nothing about where gaps and doublets appear. This can make them seem completely arbitrary, in line with Halle (1973)'s proposal that gaps are simply forms that are marked for non-insertion. As discussed by Hetzron (1975), Albright (2003), and others, however, defectiveness does not seem to be scattered randomly across inflectional paradigms.

Doublets, by definition, involve two different possibilities for the realization of a given form. Very often, each of these possibilities reflects a different generalization, as in the case of *dived* (using the default *-ed*) vs. *dove* (as in *drive*∼*drove* and *strive*∼*strove*).

Clashing generalizations seem relevant also for the distribution of gaps, though here the facts are more complex than for doublets and in some key cases are still contested in the literature. For genitive masculine singular nouns in Polish, the gaps reported in the literature seem related to the availability of two generalizations: add *-a* vs. add *-u*. Similar comments apply also for the two patterns of present-tense gaps in the Spanish verbal system discussed by Albright (2003, 2009). These gaps arise in parts of the paradigm where an alternation could apply, and Albright (2009, p. 124) notes that for the gapped verb *abolir* 'to abolish', speakers consider both the alternating form *abuelen* and the non-alternating form *abolen*. The same holds also for the genitive plural gaps in Greek studied by Sims (2006, 2009), which occur in a part of the paradigm for which the placement of stress is unpredictable: for some nouns stress shifts, while for other nouns within the same inflection class it doesn't.

The clashing generalizations that accompany a given gap need not be equally good (or bad). In fact, while speakers might volunteer (and reject) the outcomes of applying two or more clashing generalizations when asked about a given gap, they might still have a clear sense that one of those outcomes is significantly better than the others. The *forgo* gap illustrates this kind of asymmetry. The relevant generalizations in this case are the default *-ed* and the stem change that applies in *go*∼*went* and *undergo*∼*underwent*. But there is an asymmetry between the two generalizations: speakers generally dislike *forgoed* and prefer *forwent* when forced to choose. Nevertheless, speakers who are confronted with the need to produce the past tense of *forgo* occasionally mention both *forgoed* and *forwent* (before rejecting both), which suggests that both generalizations are live options.

The case of 1sg non-past gaps in Russian, discussed by Halle (1973), Albright (2009), Pertsova (2016), Gorman and Yang (2019), and others further illustrates the possibility of asymmetries between clashing generalizations. The gapped forms in this case mostly belong to the subclass of second conjugation verbs with a stem-final dental consonant. When the consonant is *t*, there are straightforwardly two different generalizations that could in principle apply. When the consonant is *d*, *s*, or *z*, however, there is just one form that is potentially licit — namely, the palatalized form — and yet gapped verbs that end in these consonants exist. However, Gorman and Yang (2019) provide evidence (building on observations by Sims 2006 and Slioussar and Kholodilova 2013) that clashing generalizations are considered by speakers also for stems ending in *d*, *s*, and *z*.

But while the correlation between gaps and clashing generalizations does not seem accidental, establishing causality is not straightforward. As the examples discussed above illustrate, there are in principle three different potential outcomes to clashing generalizations, all of them attested:

(1) Three possible (and attested) outcomes of clashing generalizations:
   a. 0 (gap): neither generalization leads to an acceptable form
   b. 1: one or the other of the clashing generalization wins over the other, resulting in a single licit choice
   c. 2 (doublet): both options are acceptable, leading to free variation

The representational and learning considerations above do not explain the connection between defectiveness and clashing generalizations, let alone predict which clashes lead to gaps, which to a single winning form, and which to doublets.

3.3 Worry 3: Speaker discomfort

A final worry, one that relates to gaps but not to doublets, is speaker discomfort (noted by Hetzron 1975, Albright 2009, and others). When asked how they would say the past tense of *forgo*, speakers don't just say that it doesn't exist. They struggle and are often unsure about whatever answer they end up giving (though the discomfort persists even in cases of clear asymmetries between clashing generalizations; see Hetzron 1975 and Pertsova 2016 for relevant discussion). This reaction is an empirical data point that should be accounted for.

It is worth noting that no similar discomfort arises for grammatical impossibilities in morphological derivation (∗*goed*) or elsewhere (∗*Ran cat the*). Deriving gaps as ungrammatical (through lack of default, as mentioned above, or by other means) does not predict speaker discomfort. Rather, it leads us to expect, incorrectly, that gaps will be similar to other cases in which unacceptability arises from ungrammaticality.[4]

4 Responses

The literature includes several ways of enriching the grammar, the learner, or both in an attempt to address one or more of the worries above.

Daland, Sims, and Pierrehumbert (2007) propose that gaps are lexicalized on an item-by-item basis (see Sims 2015 for further discussion and elaboration). In line with the learnability considerations mentioned earlier, Daland et al. propose a balanced learner. Differently from the discussion above, however, Daland et al. aim not at the learning of a morphological grammar (which in DM would include vocabulary items among other elements) but rather at the acquisition of probability distributions over the different cells in the paradigm for every stem. Gaps, on Daland et al.'s conception, are equated with cells that have very low probability. In order to account for gaps in low-frequency forms and for the skewed distribution of gaps, Daland et al. enrich the learner so as to take into consideration lexical neighborhoods, defined in a particular phonological way. This allows the learner to induce low-frequency for cells even for low-frequency stems, based on low-frequency cells in lexical neighbors.

Daland et al. 2007's account stays close to the null hypothesis in assuming the view of child-as-scientist, balancing the *a priori* goodness of a hypothesis with its fit to the data. But the target of learning that Daland et al. assume goes beyond the null hypothesis: adding a special mechanism for gaps — including the equation of gaps with low-frequency cells in a paradigm and a mechanism for acquiring distributions over cells in a paradigm — would require separate motivation, which the authors do not provide. Moreover, this additional mechanism raises nontrivial issues. For one thing, it is unclear why a low probability for a cell in a paradigm would make that cell unacceptable (an instance of the well-known problem of reducing unacceptability to low probability; see Chomsky 1957). And even if one were to add a stipulation that rules out low-probability cells in paradigms, it is hard to see why this would make speakers uncomfortable and not simply judge the relevant forms

[4]One might try to relate speaker discomfort in gaps to the absence of any straightforward good option. For run-of-the-mill ungrammaticality there are good alternatives (in a sense that can perhaps be made precise). For ∗*goed*, for example, there is *went*, and for ∗*Ran cat the* there is *The cat ran*. For gaps, on the other hand, it is often the case that no potential forms is possible. Speakers who dislike both ∗*forgoed* and ∗*forwent*, for example, cannot resort to *do*-support as a repair: ∗*John did forgo treatment* is unacceptable (see Embick and Marantz 2008 and Embick et al. 2023 for discussion). But lack of acceptable alternatives does not seem to explain speaker discomfort in gaps, given that no similar discomfort arises in cases such as island violations, where all potential choices at the relevant point in the derivation are bad and no simple repair is available (e.g., ∗*I know who you saw __ and John* and ∗*I know you saw who and John*).

as bad. That is, the third worry above remains unaddressed. And while Daland et al.'s stipulation of structuring learning according to the specific notion of lexical neighborhoods aims at addressing the first two worries (namely, low-frequency forms and distribution), it is too weak to adequately address either. For example, it does not explain why *forgo* is gapped in the past tense despite low frequency and no gapped neighbors. It also does not explain why such gaps tend to follow clashing generalizations. See Sims (2023) for further discussion of the imperfect correlation between low frequency and the actual distribution of gaps.

A different view, one that establishes a strong connection between clashing generalizations and gaps, is defended by Mendes and Nevins (2022), who propose that some gaps arise through what they refer to as *lethal competition* between two derivations within a single grammar (see also Andrews 1990, Hudson 2000, and Fanselow and Féry 2002).[5] These non-lexicalized gaps are attributed to the presence of clashing morphological expressions within the grammar, where neither expression is more specific than the other (so competition for specificity does not force a single winner). Lethal competition offers a direct handle on the distribution of gaps. It also helps in accounting for the representation and learning of gaps in low-frequency forms: the clashing generalizations are presumably acquired based on a range of different forms, so the low frequency of individual forms is irrelevant. It is much less clear why lethal competition should lead to speaker discomfort rather than just plain unacceptability. More worrying is the fact that ties within the grammar are generally taken to lead to free variation and not to ungrammaticality. If the syntax can generate both *John and Mary* and *Mary and John*, for example, then both are taken to be grammatical rather than ungrammatical. So accounting for some gaps in terms of lethal competition comes at the cost of the usual account of free variation.

Yang (2005, 2016) and Gorman and Yang (2019) attribute gaps to the *Tolerance Principle* (Yang 2016), which determines when a given process is productive. According to the Tolerance Principle, a process is productive when it has relatively few exceptions. When there are clashing generalizations, items for which just one generalization applies can count as exceptions to the other generalization. Depending on the precise counts, the number of exceptions to each generalization might be sufficiently high to prevent it from being productive. And in the absence of a productive generalization, an item that is not listed under either generalization will be a gap. This allows the Tolerance Principle to link gaps to clashing generalizations. The Tolerance Principle offers a handle also on the learning of gaps: what needs to be learned is the unproductivity of a given rule rather than a specific property of individual forms, so low-frequency forms pose no particular challenge. Speaker discomfort, on the other hand, is not explained.

A challenge to accounting for gaps using the Tolerance Principle comes from gaps with defaults. For *forgo*, for example, the Tolerance Principle needs to say that *-ed* is not productive. But it is not clear what conception of productivity would make *-ed* productive for the forms that it extends to (*google*, *fax*, *wug*) and nonproductive for *forgo*. See Pertsova (2024) for a variant of this problem for the case of 1sg nonpast verbs in Russian. This

[5]For other gaps, Mendes and Nevins (2022) propose lexicalization through lack of default.

strikes me as a nontrivial challenge.

A broader challenge to the Tolerance Principle concerns the represention of productivity. The Tolerance Principle treats productivity as a binary value that needs to be inferred (a view shared also with the Bayesian approach of O'Donnell 2015). This is a departure from the morphological view that recognizes no such value. On approaches such as DM, productivity might be used descriptively by the morphologist to convey a sense of how broadly and flexibly a given vocabulary item can be used, but it is an informal, gradient notion that can be read off a given grammar rather than a value that is written into the grammar. And if there is no productivity value that needs to be learned, there is no clear use for a special learning criterion for such a value.

A different approach is pursued by Pertsova (2005, 2016, 2024), who relates gaps not to clashing generalizations as such but rather to the alternations that are often involved in these generalizations. Pertsova adopts Steriade (1997)'s principle of Lexical Conservatism, which requires that every property of a novel form have support from a listed form. In the case of 1sg non-past form of Russian verbs ending in dental consonants, the relevant property that might need support is the palatalization that is expected to take place in the 1sg form. Pertsova (2016) notes that the past passive participial forms of a verb show the same palatalization alternation. The verb voz^jit 'to transport', for example, undergoes the same palatalization in the past passive participial forms — e.g., *vozhennij* — and in the 1sg non-past *vozhu*. Not all verbs have such passive participial forms, however, and Pertsova (2016) shows that the absence of such forms is a good predictor for gaps.

Lexical Conservatism offers a handle on the representation and learning of gaps: the principle is presumably innate, and it affects low-frequency forms that are candidates for an alternation just as easily as it does high-frequency forms. It also establishes a partial connection between gaps and clashing generalizations: alternating is one generalization and not alternating is another. However, the connection to clashing alternations is weaker than in some of the other approaches mentioned here, since Lexical Conservatism does not require non-alternation to be a grammatical possibility. Pertsova (2016) suggests that this is an advantage of her approach, since in the case of Russian 1sg non-past forms, not alternating does not seem to be a licit option for verbs ending in dental consonants. Recall, however, that Gorman and Yang (2019) argue that clashing generalizations are present also in the case of Russian 1sg gaps, appearances to the contrary notwithstanding. Finally, as in most of the other proposals above, speaker discomfort remains unaddressed under Lexical Conservatism.

Accounting for gaps in terms of Lexical Conservatism raises several concerns. For one thing, it is unclear why lack of supporting forms might lead to gaps to begin with (see Sims 2017). Steriade (1997)'s argument for Lexical Conservatism involves cases where lack of support for an alternation from a different form corresponds to non-alternation, but the result is judged acceptable. Why should Russian 1sg gaps be different, with lack of a supporting alternating form resulting in unacceptability rather than in nonalternation?

One could attempt to address this worry, as Pertsova 2016 does, by making the alternation requirement very strict for the relevant Russian verbs, so that if they alternate they will violate Lexical Conservatism and if they don't they will violate the alternation requirement. This approach requires a further mechanism to ensure that when all output possibilities violate requirements that are sufficiently strict, the result is ungrammatical. Pertsova 2016 refers to this as a threshold approach and discusses a way in which it can be implemented within a particular constraint-based system. As far as I can see, a threshold mechanism would serve the sole purpose of ruling out winners within an architecture that otherwise allows winners to surface. In addition to being an otherwise unmotivated complication, a threshold approach faces an empirical challenge: it needs to make both Lexical Conservatism and the alternation requirement sufficiently important to ensure that violating either would prevent a form from passing the relevant threshold. Both parts are necessary for the threshold approach, but neither seems straightforward. Speakers extend alternations to novel forms, including in Russian (Tomas et al. 2017). This suggests that Lexical Conservatism, to the extent that it holds, is not sufficiently important to block an output. And the importance of an alternation requirement is challenged by Slioussar and Kholodilova (2013)'s evidence that various recent borrowings from English into Russian surface without alternation in the 1sg nonpast.

Even if one provides a way to derive unacceptability (and not just nonalternation) using Lexical Conservatism and a threshold approach, some gaps remain unexplained. In the Russian case, Pertsova (2016, p. 17) notes that several defective verbs do, in fact, have alternations elsewhere. These gaps would then require a different explanation.

Albright (2003, 2009) accounts for gaps by incorporating uncertainty into the grammar. The grammar is taken to include connections between different parts of different paradigms. These connections are inferred by learners using a particular method. Gaps are taken to be cases in which speakers end up with low confidence in a given trans-derivational connection. Albright 2003's proposal offers a handle on all three of the empirical worries mentioned earlier: gaps in low-frequency cases are expected, since low frequency is compatible with low confidence and can in fact contribute to it; clashing generalizations are expected to be relevant, since they serve to increase uncertainty; and relying on a notion of confidence, even if it is within the grammar, suggests a source of unacceptability that is different from usual cases of ungrammaticality.

I believe that speaker uncertainty is indeed central to an account of gaps. In this key regard, the present proposal will be in agreement with Albright (2003, 2009)'s account. But adopting trans-derivational dependencies as grammatical objects is a significant architectural departure from the null hypothesis, as is the learning method that Albright uses. As noted by Pertsova (2016, p. 10), this choice of learning method is not accidental: for the proposal to yield gaps, the learner needs to ignore broad, robust generalizations. This rules out learning approaches such as the balanced learning advocated here. Moreover, Pertsova (2016, p. 11) notes that the specific learning approach used by Albright (2003, 2009) does not seem to make the right predictions for the case of Russian 1sg nonpast gaps.

Another departure from the null hypothesis concerns the representation of uncertainty within the grammar itself. In what follows I will keep Albright (2009)'s proposal that gaps are cases of uncertainty, but I will treat the uncertainty to be about the correct hypothesis rather than being within the grammar. I will show that this idea fits naturally within common views on representation and learning.

5 Proposal

As reviewed in the previous section, different proposals in the literature take gaps and doublets to require nontrivial departures from the null hypothesis. Such departures might be merited if the empirical gains from these departures are sufficiently significant to justify the costs incurred in complicating the theory. As I tried to show, this is not the case for the proposals reviewed: the theoretical complications are problematic not just conceptually but also empirically. In particular, none of the existing proposals adequately addresses the three worries regarding representations and learning, distribution, and speaker discomfort.

Common to the proposals reviewed in the previous section is the assumption that gaps and doublets should be derived within individual grammars. This assumption is unwarranted, and the present section shows that once this assumption is abandoned, gaps and doublets become much less puzzling.

In rejecting the requirement that gaps and doublets be derived within individual grammars I will rely on two assumptions discussed in the literature and mentioned briefly in the introduction:

(2) a. MULTIPLE GRAMMARS WITHIN A HYPOTHESIS: a hypothesis is not necessarily a single grammar; it can consist of multiple grammars, which may have specific contexts of use (register) but can also be in free variation

b. UNCERTAINTY ABOUT THE CORRECT HYPOTHESIS: acquisition is sometimes incomplete, in the sense of leaving human learners with uncertainty about various aspects of the winning hypothesis

Assumption (2a) is supported by bilingualism and register and has been argued for by Roeper (1999) and others. Assumption (2b) is at the heart of the view of the child as scientist already discussed. I take both of them to belong to the null hypothesis and will use them in my account of gaps and doublets. Before presenting the account I will need to be more explicit about the effects of the uncertainty in (2b).

5.1 A knowledge norm on forms

How does uncertainty between hypotheses relate to judgments? One could imagine, for example, that in the face of uncertainty, speakers would accept all leading possibilities. One could also imagine that speakers would always opt for the forms favored by the leading hypothesis. Neither of these choices would derive gaps, however, and in section 1 I suggested a third choice, according to which using a form requires knowing that it is grammatical according to the correct hypothesis. I will state this choice as follows:

(3) KNOWLEDGE NORM ON FORMS: to use a given form x, the speaker needs to know that x is derivable according to the correct hypothesis

The knowledge norm on forms is modeled after the knowledge norm of assertion, defended by Williamson (1996, 2002) and discussed in much subsequent literature (see Benton 2024 for a recent overview). In the domain of meaning, it has been argued that asserting something requires knowing that it is true and not just believing that it is likely. If I haven't checked the weather outside I can neither assert that it's raining nor that it's not raining, regardless of how likely I find each possibility to be. If forced to assert one or the other, I would presumably choose the likelier option, but the assertion would still be infelicitous. The current proposal, then, says that not just the content of assertions but also the forms used in utterances are subject to a requirement of knowledge.[6]

The knowledge norm on forms is no more stipulative than other choices that could be made, such as license to use forms that are grammatical according to some high-ranking hypothesis or license to use forms that are grammatical according to the highest-ranking hypothesis. And in light of the broader context that includes the knowledge norm on assertions, the knowledge norm on forms strikes me as more natural than those other theoretical choices. But I don't know if this choice might reduce to more basic considerations.

Let us see how the null hypothesis and the knowledge norm on forms can derive gaps and doublets, starting from the latter and from assumption (2a). Adult speakers have at least one grammar, but typically many more. This is most clearly the case for multilingualism but can also be seen in the ability of speakers to switch between registers. More broadly, as discussed by Roeper (1999), speakers can be seen as having many different grammars as part of the adult state. For acquisition, then, the hypotheses that the child considers can include *sets* of grammars rather than being limited to just a single grammar. It is often the case that all the grammars in the winning hypothesis will agree. For example, the different grammars in the winning hypothesis of English speakers will agree that the past tense of *walk* is *walked*. But it is conceivable that the winning hypothesis will contain two grammars that disagree about a given form. For example, $G1$ might derive *dived* and $G2$ might derive *dove*. In this case, and unless the grammars differ in when they can be used (as in the case of register), both forms will be possible. Note that while there is competition between (the processes behind) *dived* and *dove* within $G2$ (with *dove* as the sole winner), there is no such competition within $G1$ (where *dove* is simply not derived); the observed variation between *dived* and *dove* arises not within either of the grammars but from the multilingual configuration. Of course, having both $G1$ and $G2$ in H is more complex than having just one, which on common assumptions about simplicity (as mentioned above) requires more evidence for acquiring H than for acquiring a hypothesis that has just $G1$, for example, or just $G2$. Note, however, that since the differences between $G1$ and $G2$ are likely to be very minor, a sensible representation of $H = \{G1, G2\}$ will take advantage of this overlap and not be much longer than either $H1 = \{G1\}$ or $H2 = \{G2\}$. And encountering both *dived*

[6]An obvious worry is what counts as knowledge for purposes such as norms on speaker behavior. In what follows I simply assume that whatever notion is relevant for assertion is relevant also for (3).

and *dove* at above-noise levels in the primary linguistic data should suffice for a rational learner to justify adopting H (rather than $H1$ or $H2$).[7]

Turning to assumption (2b) — and to gaps — consider what happens when the learner is uncertain about a given aspect of the winning hypothesis. For example, the learner might not be sure what past tense is licensed for *forgo* by the correct hypothesis: perhaps they find it equally likely that the correct hypothesis licenses *forgoed* and that it licenses *forwent*, or perhaps they find one of the possibilities much likelier than the other but are still not certain. Why should speakers be uncertain? One possibility is that learners entertain the hypothesis of a VI for $\sqrt{\text{FORGO}}$ in the past tense that involves stem suppletion analogously to $\sqrt{\text{GO}}$. A more plausible possibility, pointed out to me separately by Adam Albright and Jonathan Bobaljik, is that the uncertainty is about whether *forgo* is decomposed into *for+go* or not: in the former case, the irregular *forwent* should surface (similarly to *undergo*$\sim$ *underwent*) and in the latter case it shouldn't. A similar explanation is offered by Nikolaev and Bermel (2022, p. 586) for the defectiveness of compound verbs such as *troubleshoot*. Regardless of the source of uncertainty, the speaker is not in a position to satisfy the knowledge norm on forms with either *forgoed* or *forwent*, and the form is gapped.[8]

5.2 Back to the three worries

With the knowledge norm on forms in place, we can now return to the three worries about gaps and doublets: representation and learning; distribution; and speaker discomfort.

5.2.1 Representation and learning

The worry about gaps was that learning the lack of a default requires sufficient indirect negative evidence. This was problematic for low-frequency cases. The current proposal addresses this worry by attributing some gaps to speaker uncertainty about the correct hypothesis. As in Albright (2003, 2009)'s account, low frequency is perfectly compatible with uncertainty and may in fact contribute to it, so there is a path for gaps to arise even in low-frequency cases. The worry about doublets was that a specific form (e.g., *dove*) might be expected to block a default (e.g., *dived*). While a solution existed — using hyper-specific versions of *-ed* and ablaut — it did not seem correct. The current proposal addresses this worry by allowing doublets to arise from the existence of multiple grammars within the correct hypothesis. With suitable representations, splitting the grammar for, e.g., *dived/dove*

[7]For doublets where neither process is more specific than the other, representation within an individual grammar is still possible. And since it is the representationally simpler option, it will be acquired, and no split will arise.

[8]Note that speaker uncertainty can be discussed intensionally, as above, in terms of speakers considering given choice points with respect to a winning hypothesis. A different way to discuss speaker uncertainty is extensionally, in terms of a set of leading hypotheses that speakers consider, where each hypothesis corresponds to a full set of choices for the various choice points. I will sometimes use the latter, extensional perspective below, but only for presentational convenience. I do not wish to imply that speakers actually extensionalize their beliefs about different hypotheses.

can be much simpler to encode than the hyper-specific entries needed within an individual grammar. A balanced learner would therefore adopt the former.

5.2.2 Distribution

The worry about distribution was that gaps seemed correlated with the presence of clashing generalizations but that the correlation was imperfect: depending on the particular example we may find no acceptable forms (gaps), a single form (the typical case), or multiple forms (doublets).

Clashing generalizations lead to gaps when speakers have not seen sufficient evidence to know which of these generalizations apply according to the correct hypothesis. For *forgo*, relevant evidence could be instances of the actual forms *forgoed* or *forwent* at above noise levels, or it could be sufficient indirect negative evidence to support a lexicalized gap. Or it could be various kinds of evidence telling the learner whether *forgo* decomposes into *for+go*, as mentioned above. But many speakers presumably do not have access the critical evidence and so are left without knowing what the correct hypothesis says in this case. They might have a preference between a hypothesis that derives *forgo* and one that derives *forwent*, but this preference does not amount to knowledge.

Clashing generalizations lead to a single form when one of these generalizations is taken to be more specific than the rest according to the correct hypothesis or when the correct hypothesis is taken to make just one generalization applicable to a given item.

Finally, doublets arise when neither of two generalizations is taken to be more specific than the other according to the correct hypothesis or when learners are exposed to sufficiently many instances of both generalizations applying to a given item. As discussed above, it is the latter case that leads to splits into multiple grammars. For *dive*, speakers presumably hear both *dived* and *dove*. If this evidence is above noise levels, speakers will know that both are grammatical according to the correct hypothesis. This means that the correct hypothesis has two grammars (since a single grammar will not accommodate the derivation of both *dived* and *dove*; the specific *dove* will block the default *dived*).

The current proposal explains the appearance of trans-derivational constraints without committing to a grammatical formalism that allows for the statement of actual trans-derivational constraints. Suppose that there is a near-tie between the two leading hypotheses, $H1 = \{G1\}$ and $H2 = \{G2\}$ that conflict regarding a given form X. $G1$ might express X as x_1 while $G2$ expresses it as x_2. In the case of Russian 1sg nonpast forms, for example, $G1$ might derive x_1 by applying palatalization to the UR of X while $G2$ does not apply palatalization. Without further information, speakers would have a gap for X, arising from their inability to choose between $H1$ and $H2$, which results in an inability to tell whether the correct form is the palatalized x_1 or the unpalatalized x_2. Suppose, however, that there is another part of the paradigm that is a good predictor of whether palatalization should apply. E.g., palatalization in the passive participial part of the paradigm might predict palatalization in the nominative singular. In that case, forms in which plural parts of the paradigm are (a) observed, and (b) informative about palatalization will help the child break the tie and avoid having a gap. The result is something that looks superficially like trans-derivational constraints. But this appearance of trans-derivational constraints does not imply the existence of such constraints within the grammar.

Finally, note that the current proposal predicts that the distribution of gaps and doublets will vary to some extent between speakers and over generations. A child who hears *forgoed* a few times, even if this is a production error of speakers for whom this form is gapped, can acquire *forgoed*. If they also happen to hear *forwent* (again, possibly as production errors by speakers for whom the form is gapped) they may acquire a doublet. The current proposal also predicts that gaps will occasionally arise by accident and then entrench themselves and that doublets will occasionally disappear. Consider a case in which the child needs to choose between the correct $H1 = \{G1\}$ and a nearly identical $H2 = \{G2\}$, where $G1$ and $G2$ differ with respect to just one form, X: $G1$ expresses X as x_1 while $G2$ expresses it as x_2. If the child happens to hear x_1 for X they might converge on $H1$. But if the child happens not to hear x_1, which is likely to happen if X has low frequency, they will have uncertainty, and consequently a gap — one that was not there for the previous generation. The above predicts that multiple generalizations can lead to new gaps in low-frequency cases. Note that this maintains a clear distinction between plain low-frequency forms and actual gaps. As to the elimination of doublets, suppose that the correct hypothesis is not $H1$ but rather $H = \{G1, G2\}$. If the child hears just x_1 at above noise levels, they will be justified in acquiring the simpler $H1$ rather than the correct H, losing the doublet of the previous generation.

5.2.3 Speaker discomfort

The current proposal addresses the worry about speaker discomfort by treating the relevant gaps as cases where no form satisfies the knowledge norm. For these gaps, speakers don't know whether a given form is right, which is different from familiar cases of unacceptability where speakers know that a given form is not right.

Note that there is a parallel here with presuppositions. Fox (2008) and George (2008) argue that presupposition failure involves an unknown classical truth value, in line with the Strong Kleene trivalent logic (Kleene 1952). If John has no children, the sentence *John's children are bald* has a classical truth value — either true or false — but we don't know what it is. The discomfort that speakers experience in asserting the sentence in such a context might then be similar to their discomfort in asserting that it's raining when they find it highly likely that it's raining but don't know that it's raining. In both cases, the discomfort is reduced to the knowledge norm of assertion. The present proposal connects this discomfort to speakers' discomfort with gaps.

The parallel with presupposition might go further than that. Fox (2008) and George (2008) argue that treating presupposition failure as uncertainty about a classical truth value makes it possible to predict the pattern of presupposition projection from within complex sentences. Simplifying considerably, presupposition projection is reduced to uncertainty projection. For example, if A is false and the truth value of B is unknown (either true or false but we don't know which), then $A \wedge B$ is false — we can determine that without knowing the truth value of B. $A \vee B$, on the other hand, has an unknown truth value. And if A is true and the truth value of B is unknown, $A \wedge B$ is unknown while $A \vee B$ is true.

On the current account, we should expect to find something similar with uncertainty gaps. Consider a gap that arises from uncertainty between two hypotheses, $H1$ and $H2$, the former licensing x_1 and not x_2 for a given X and the latter licensing x_2 and not x_1. In a context in which speakers do not need to actually use the form, the knowledge norm on forms is not at stake for X, and the sentence can be acceptable. Any process that obliterates the diffence between x_1 and x_2 should in principle serve this purpose. An obvious process of this kind is phonological deletion, and building on Mendes and Nevins (2022), I believe that the prediction of the current account is correct.

If a structure includes the past of *forgo* but is phonologically null (for example, due to VP ellipsis), then speakers might feel comfortable: while they might know know whether the correct hypothesis licenses *forgoed* or *forwent*, they might still know that it licenses some form, and whichever form it is, the phonological output in an ellipsis context will sound the same (namely, null). In such a case, the knowledge norm on forms is satisfied, and speakers might be willing to accept the relevant form.[9,10]

6 Conclusion

I proposed that gaps can arise from uncertainty about what the correct hypothesis says and that doublets can arise from two grammars within the correct hypothesis. Uncertainty about the correct hypothesis and multiple grammars within a hypothesis are assumed quite broadly and can be taken to be part of the null hypothesis. My account of gaps relied also on a knowledge norm on forms, which is perhaps not part of the null hypothesis but seems no more stipulative than other choices of how to handle uncertainty. In light of the connection to the knowledge norm on assertion, the present choice may in fact be more natural than some other prominent possibilities, as discussed above. Be that as it may, the present account avoids the rather major departures from the null hypothesis that have been entertained in the literature, and it addresses the three worries about (a) representations and learning, (b) distribution, and (c) speaker discomfort in a more satisfactory way than other proposals.

[9] As pointed out to me by Jonathan Bobaljik (p.c.), uncertainty can in principle be eliminated through other means as well. In particular, he notes that certain cases in the syntactic literature that have been argued to require matching but can be rescued by syncretism might be reanalyzed in terms of uncertainty and its elimination. I will not be able to examine these cases in any detail within the present paper.

[10] Mendes and Nevins (2022)'s take on salvation by deletion is different. They suggest that gaps that are due to PF ill-formedness (specifically, to a missing phonological exponent) can be rescued by phonological deletion while gaps that are due to semantic ill-formedness (specifically, to a missing encyclopedic entry) cannot be rescued in this way. It is not clear to me that deletion should indeed undo PF ill-formedness, and on the current account it is not required to do so: deletion simply makes it possible to avoid a pragmatic problem (namely, a violation of the knowledge norm on forms) and does not repair ungrammaticality.

The present account is stated in terms of general principles of grammar and cognition. Why, then, does the literature on gaps seem to focus specifically on (inflectional) morphology? To some extent, this focus makes sense. Morphological processes vary between items, so for a given form, speakers can easily be left with gap-inducing uncertainty. Phonology, syntax, and compositional semantics, on the other hand, tend to apply across-the-board. There will therefore be no occasion for two grammars to vary in whether, e.g., a given determiner composes with a given noun or whether topicalization can apply to the resulting noun phrase.

But there are places where uncertainty about items (perhaps because of morphological considerations) might give rise to uncertainty that seems to go beyond morphology. I will briefly mention three potential cases of this kind but will not attempt to analyze them in any detail within the present note.

One suggestive example was presented already by Hetzron (1975), who notes that Russian has configurations in which the preposition required by the verb clashes with the preposition required by the noun. In these configurations, a gap arises. Another example, pointed out to me by Jonathan Bobaljik (p.c.), concerns the possessive form of coordination with personal pronouns, discussed by Herriman 1999 and Payne 2011. For at least some English speakers, ??*Ann and my paper*, ??*Ann's and my paper*, and ??*Ann and me's paper* are all degraded, with no acceptable way to form the relevant possessive. This might be amenable to an analysis in terms of uncertainty: speakers might not know what the correct hypothesis says for such cases. And Nina Haslinger (p.c.) pointed out to me a syntactic gap discussed by Koopman (1995), Fanselow and Féry (2002), Vikner (2005), and others for certain complex verbs in German and Dutch, such as German *bergsteigen* 'to mountain-climb'. In contexts where the verb should move to V2 position, speakers have a gap. This can be attributed to an uncertainty about whether the verb splits: if it does, the second half moves to V2 while the first half is clause final; otherwise, the whole verb moves to V2. So decompositional uncertainty of the verb might lead to a syntactic gap.

The three cases above are suggestive of uncertainty gaps beyond the strictly morphological domain, a phenomenon that is predicted by the current account but puzzling for the accounts surveyed in section 4, all of which tie gaps to specific properties of morphological paradigms. Of course, a proper analysis would require more than the brief overview provided here.

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